Lecture 07:

Cross Validation & Bias-Variance Trade-off

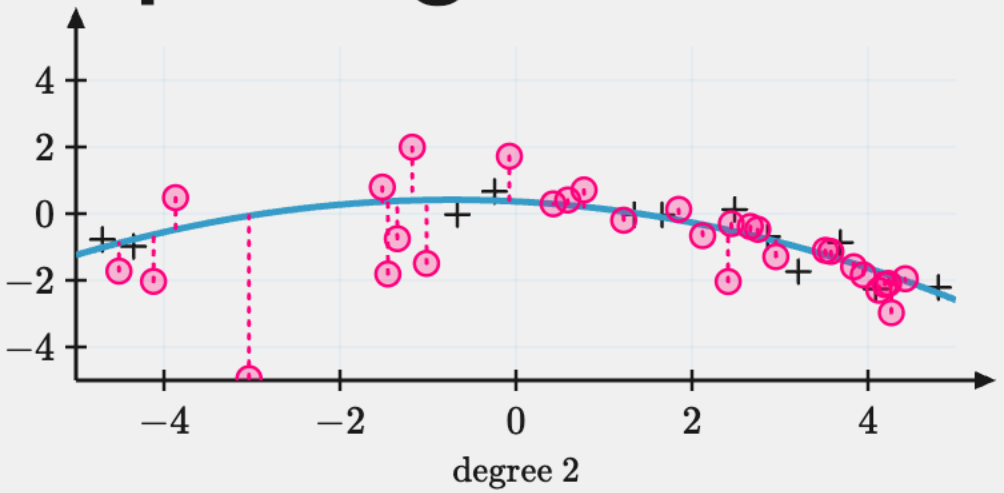
Matthew Wicker

27 October 2026

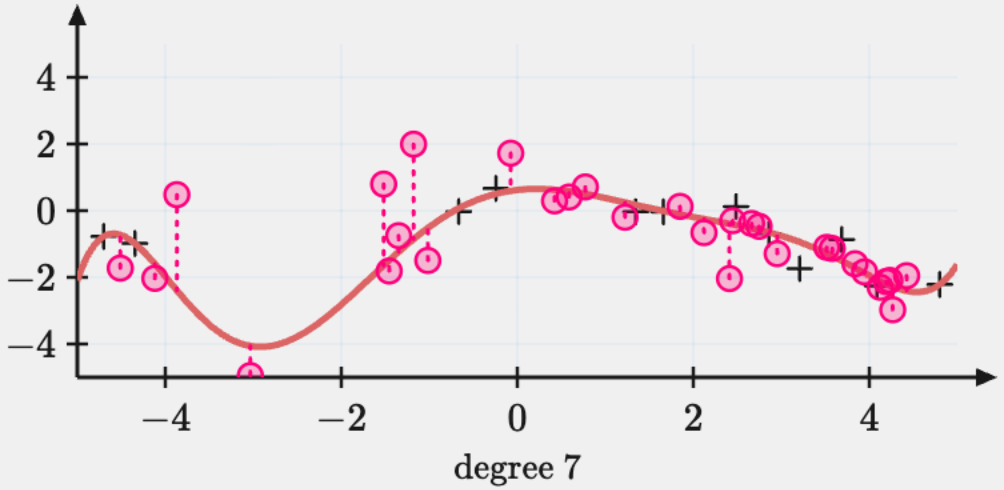
Coursework: Kryptonite v3.0

- The coursework, *Kryptonite v3.0*, is released today and is due in 3 weeks: Tuesday 17 November.
- In groups of 3 to 4: read the Kryptonite-3.0 mock paper, explore its challenge datasets with a valid experimental design, and write a response paper that refutes its claims.
- Required: adapt its foundation model, KryptoNet-40, with LoRA, explaining LoRA and counting its operations.
- Submit a 3–6 page report in conference format, code that reproduces every figure, and your predictions for the hidden test sets.
- Everything is at github.com/matthewwicker/Kryptonite_v3.0

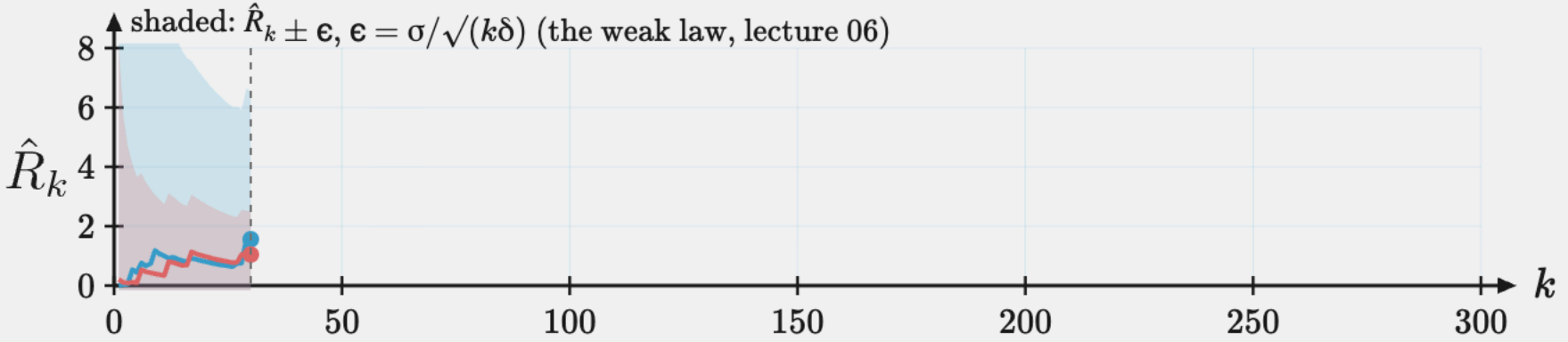
Comparing models on a test set



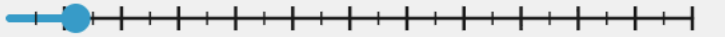
$\hat{R}_k = 1.56 \pm 5 \;\; (k = 30)$



$\hat{R}_k = 1.05 \pm 1.46 \;\; (k = 30)$ **lower**



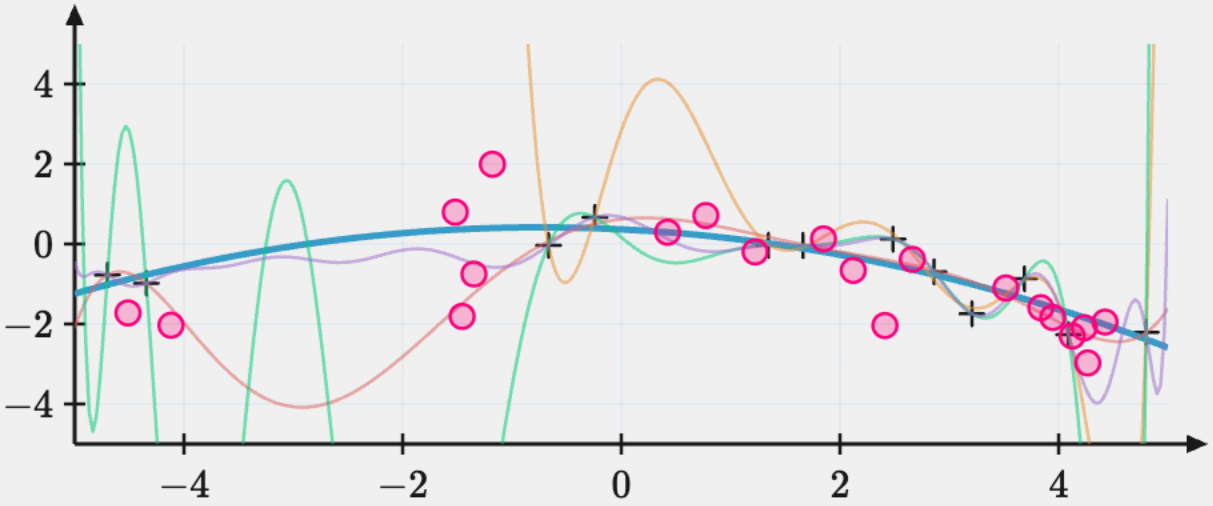
$k = 30$



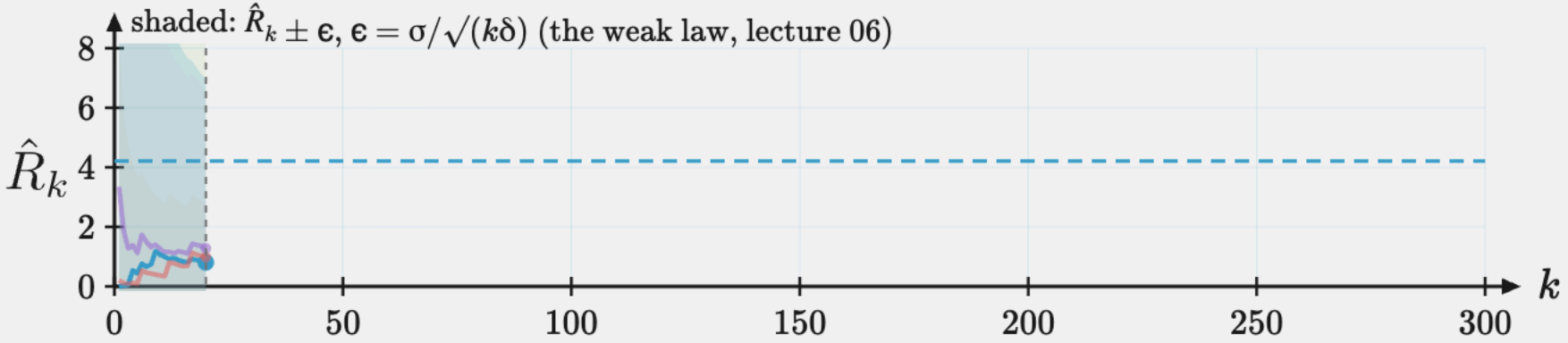
$\delta = 0.10$



Picking the best of many models



degree 2: 0.81 ± 6.13 ← picked
degree 7: 0.98 ± 1.78
degree 10: 7030.12 ± 2.0×10^5
degree 15: 36.83 ± 58.6
degree 20: 1.28 ± 5.27
true risk of the pick: **4.21** (dashed)
over 400 trials of 20 points, the pick's estimate is on average **0.13** below its true risk



$k = 20$

50 100 150 200 250 300

$\delta = 0.10$

0.1 0.2 0.3 0.4 0.5

Hyper parameter selection *is* learning

$$\lambda^\star, t^\star = \underset{\lambda, t}{\operatorname{argmin}} \min_{\theta} \mathbb{E}\left[R\left(f^{M(\theta', \lambda, t)}(\mathbf{x}), \mathbf{y}\right)\right]$$

outer level: the hyper-parameters λ, t · inner level: the parameters θ

Here is a traditional set up with a number of steps we will learn for (gradient descent) and a regularization strength λ.

You can look at the process of learning as having two levels.

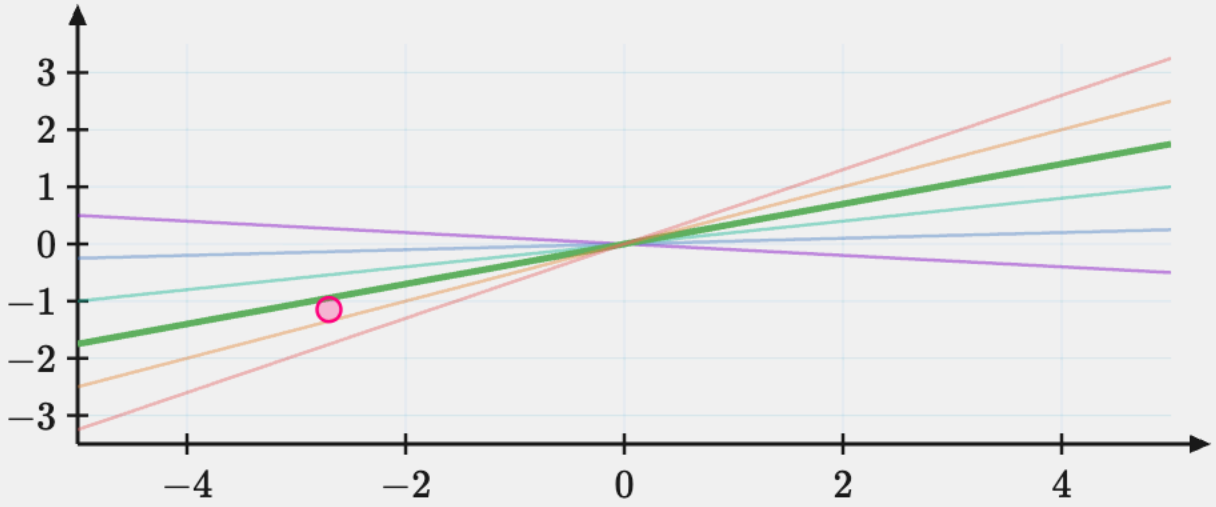
Hyper parameter selection *is* learning

$$\lambda^\star, t^\star, \theta_1^\star = \underset{\lambda, t, \theta_1}{\operatorname{argmin}} \; \underset{\theta / \theta_1}{\min} \; \mathbb{E}\left[R\left(f^{M(\theta', \lambda, t)}(\mathbf{x}), \mathbf{y} \right) \right]$$

But from the perspective of overfitting, this separation is not clear – the separation is only a conceptual one.

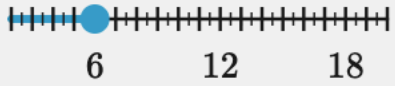
Imagine someone makes the arbitrary choice to make their bias a hyper-parameter... you can see how this leads to overfitting!

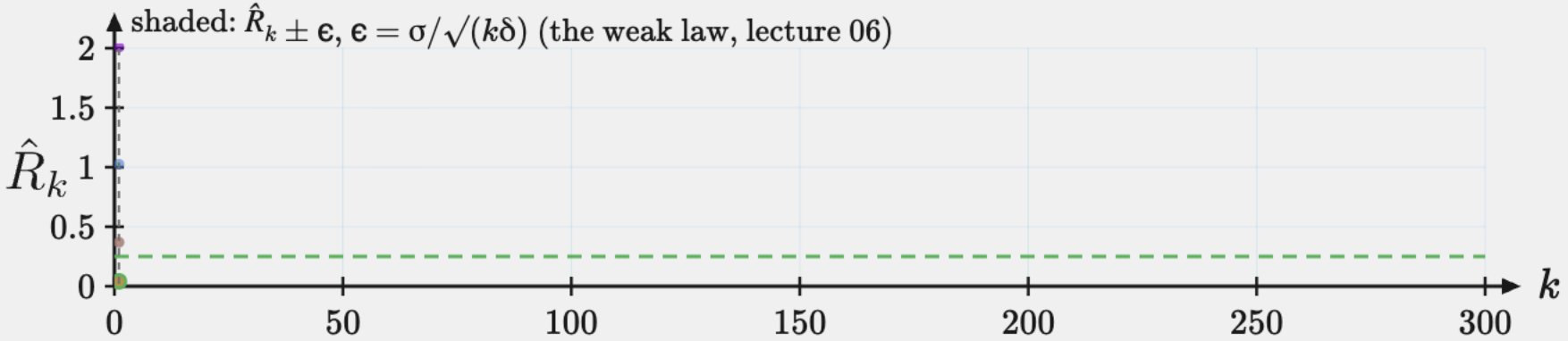
Picking a parameter on the test set



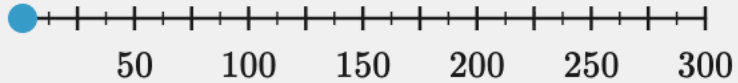
6 values of θ from −0.1 to 0.65, $f(x) = \theta x$
picked: $\theta = 0.35$, $\hat{R}_k = 0.04 \pm 1.12$
true risk of the pick: **0.25** (dashed)
over 400 trials of 1 point, the pick's estimate is on average **0.63** below its true risk

$d = 6$

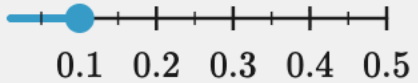




$k = 1$



$\delta = 0.10$



Can we repair our guarantee?

Yes!

The field of statistical learning theory studies guarantees on learning algorithms, and the simplest tool we can present here is the **union bound**, which repairs our guarantee.

Guarantees must be stated *a priori*

A valid guarantee on the risk must be stated *a priori*, that is, before the data is seen.

Let's consider a comparison of '*d*' many models, as when we picked the parameter θ. If we know, *a priori*, that the risk estimates hold simultaneously, then we have a valid estimate:

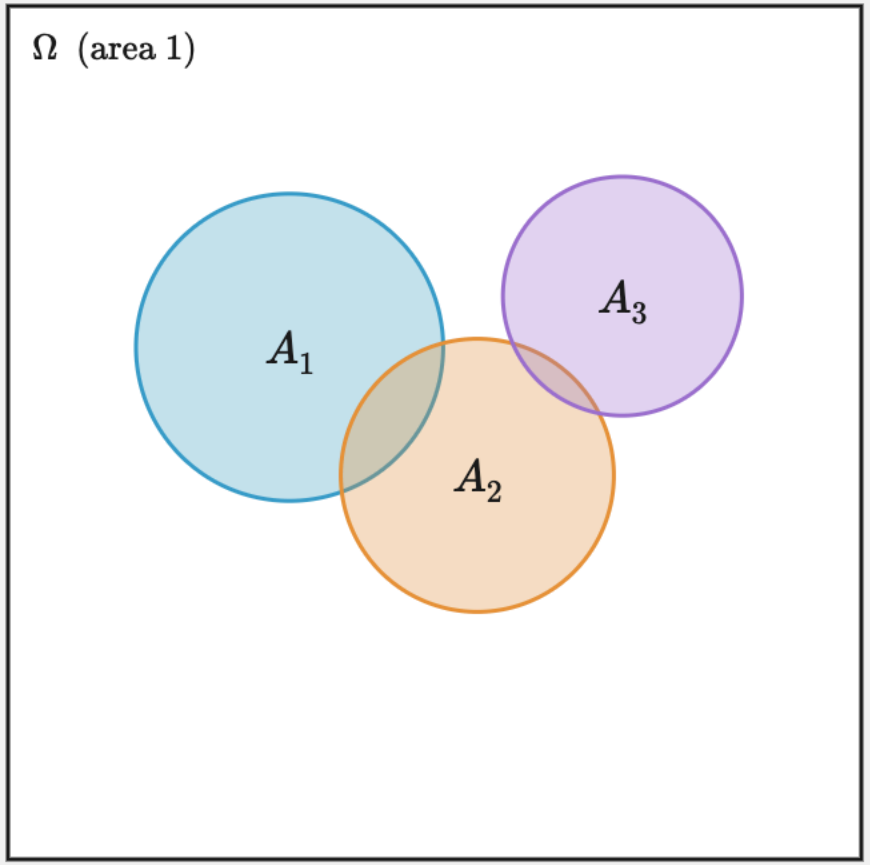
$$P\left[\forall j \in \{1, \ldots, d\} : \left|R(\theta_j) - \hat{R}_k(\theta_j)\right| \leq \epsilon\right] \geq 1 - \delta$$

Union bound in probability

The union bound allows us to reason about union events by making a "worst case" assumption that the events are pairwise disjoint:

$$P\left(\bigcup_{i=1}^{n} A_i\right) \leq \sum_{i=1}^{n} P(A_i)$$

The union bound, live



$P(A_1 \cup A_2 \cup A_3) = 0.231$ (the area covered)

$P(A_1) + P(A_2) + P(A_3) = 0.244$

the sum counts the overlaps more than once: 0.013 too much

$radius = 0.160$



Using the union bound on generalization

Our *a priori* guarantee for *d* models compared on the same test points:

$$P\Big[\,\forall j \in \{1,\ldots,d\} :\ \big|R(\theta_j) - \hat{R}_k(\theta_j)\big| \le \epsilon\,\Big] \ge 1-\delta$$

Equivalently, *some* estimate misses by more than ϵ with probability at most δ:

$$P\Big[\,\exists j \in \{1,\ldots,d\} :\ \big|R(\theta_j) - \hat{R}_k(\theta_j)\big| > \epsilon\,\Big] \le \delta$$

the complement: some estimate misses its risk by more than ϵ

$$= P\Big[\bigcup_{j=1}^{d} \big\{\big|R(\theta_j) - \hat{R}_k(\theta_j)\big| > \epsilon\big\}\Big]$$

"for some *j*" is the union of the *d* events

Using the union bound on generalization

The union bound, for the events A_j that model j's estimate misses:

$$P\Big(\bigcup_{j=1}^{d} A_j\Big) \le \sum_{j=1}^{d} P(A_j), \qquad A_j = \big\{\big|R(\theta_j) - \hat{R}_k(\theta_j)\big| > \epsilon\big\}$$

$$P\Big[\bigcup_{j=1}^{d} \big\{\big|R(\theta_j) - \hat{R}_k(\theta_j)\big| > \epsilon\big\}\Big] \le \sum_{j=1}^{d} P\big(\big|R(\theta_j) - \hat{R}_k(\theta_j)\big| > \epsilon\big)$$

$$\le \sum_{j=1}^{d} \frac{1}{4k\epsilon^2}$$

each term by the weak law (lecture 06): the losses are in [0, 1], so $\sigma^2 \le 1/4$

$$= \frac{d}{4k\epsilon^2}$$

d equal terms

Using the union bound on generalization

We want this to be at most δ, and solve for ϵ:

$$\frac{d}{4k\epsilon^2} \leq \delta \Longrightarrow \epsilon^2 \geq \frac{d}{4k\delta} \Longrightarrow \epsilon \geq \sqrt{\frac{d}{4k\delta}}$$

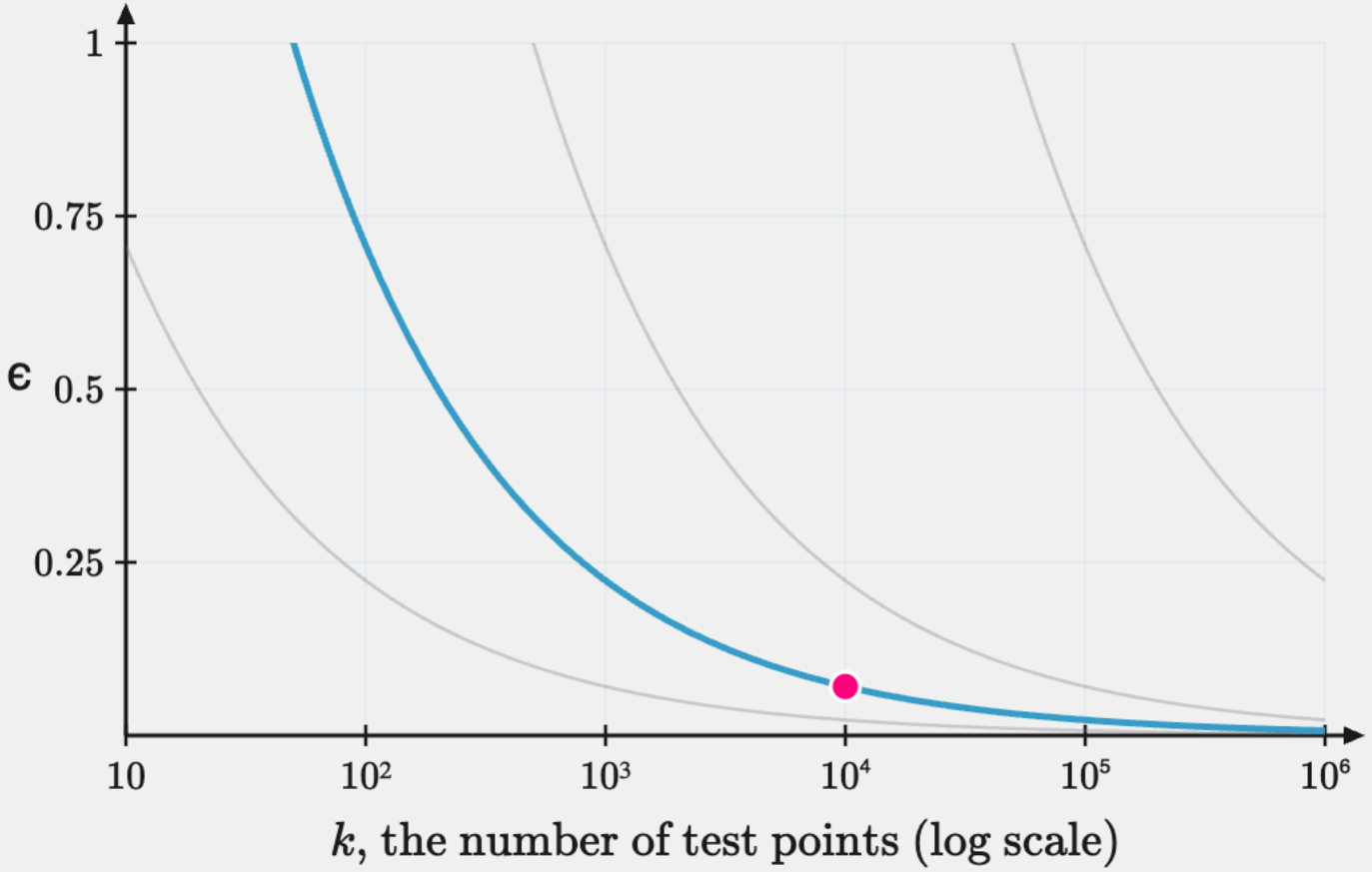
multiply by $4k\epsilon^2$, divide by $4k\delta$ square roots (both positive)

So, stated *a priori*, before the data are seen:

$$\text{with } \epsilon = \sqrt{\frac{d}{4k\delta}}: \quad P\left[\forall j \in \{1, \ldots, d\}: \left|R(\theta_j) - \hat{R}_k(\theta_j)\right| \leq \epsilon\right] \geq 1 - \delta$$

a guarantee for every model at once, whichever we pick: the curves on the next slide

The generalization bound, live

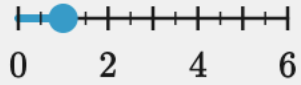


$d = 10$ models, $\delta = 0.05$

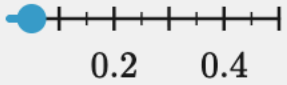
with $k = 10{,}000$ test points: $\epsilon = 0.071$

for $\epsilon = 0.05$: $k \geq 20{,}000$

$log_{10}\, d = 1.0$



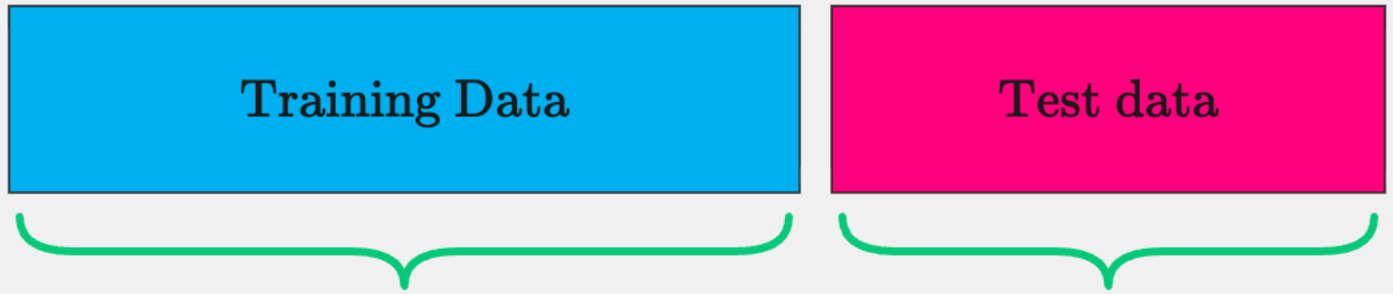
$\delta = 0.05$



$log_{10}\, k = 4.00$



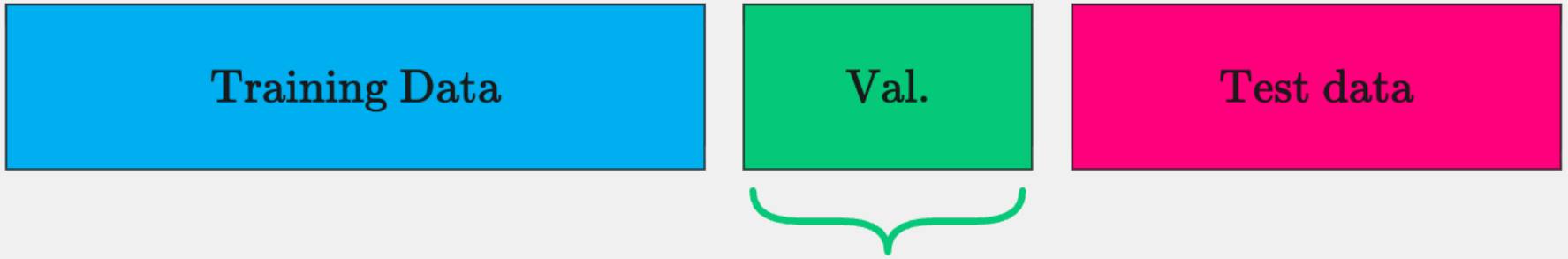
So how do we choose hyper-parameters?



Which decisions you make are trivial because they are all universal function approximators

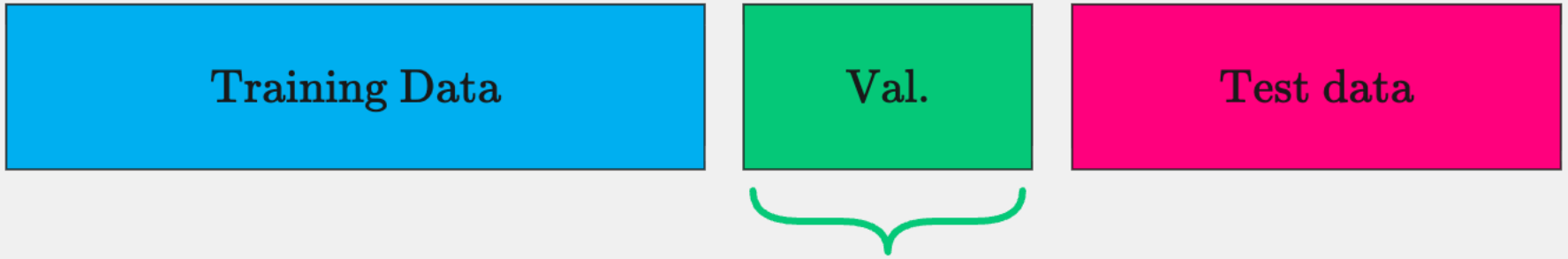
Leads to invalid risk estimates

So how do we choose hyper-parameters?



We can introduce the idea of a “validation” dataset that we make hyper-parameter choices with, and then we can still use our test data after to understand the true risk

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We can introduce the idea of a “validation” dataset that we make hyper-parameter choices with, and then we can still use our test data after to understand the true risk

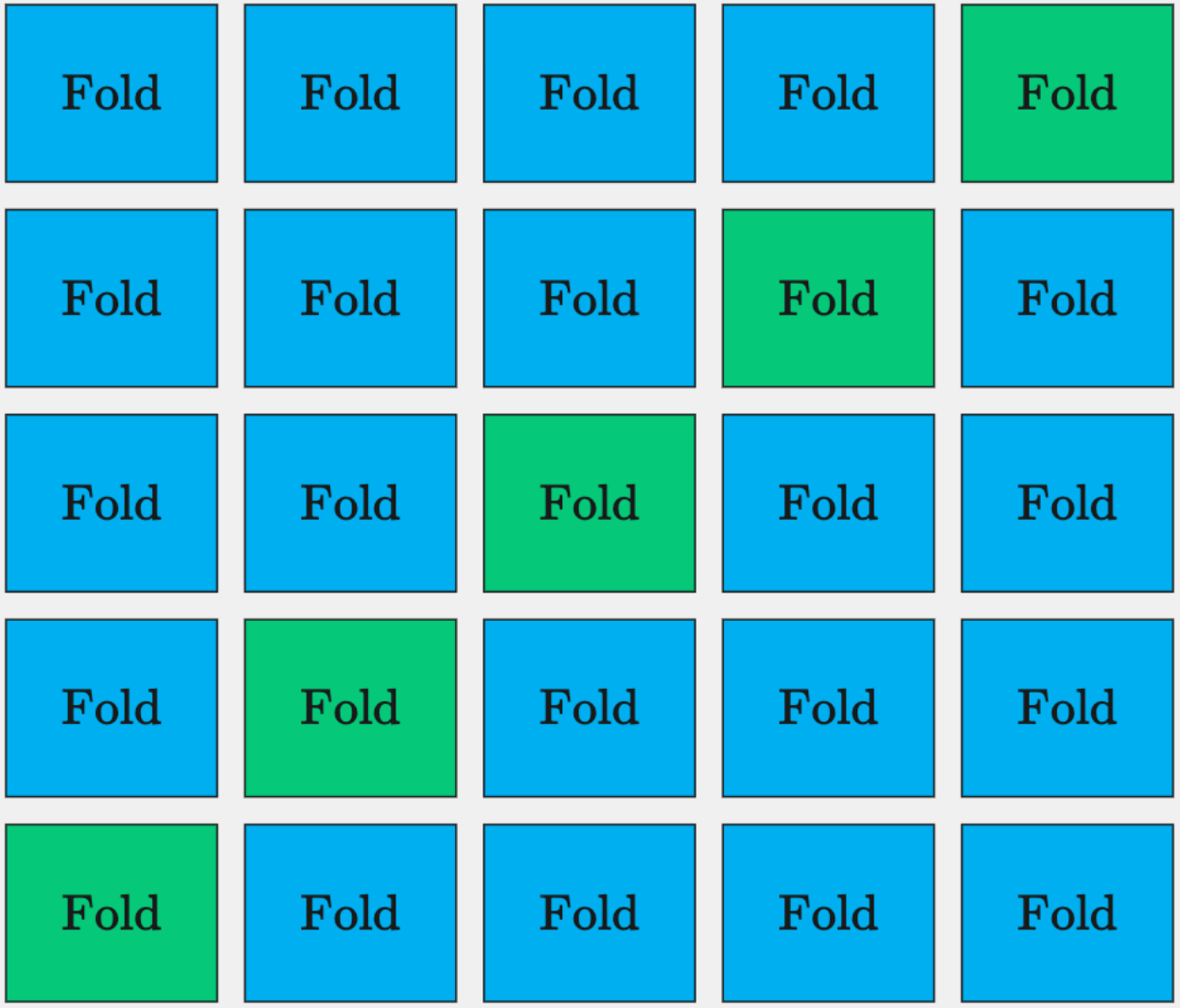
But now we have split our data into many sets – and if the validation set is small, it may be a bad estimator! (low ‘*k*’)

K-fold cross-validation



K-fold cross validation splits our training data into '*k*' different folds. Then, we measure how well the model performs using a held out fold.

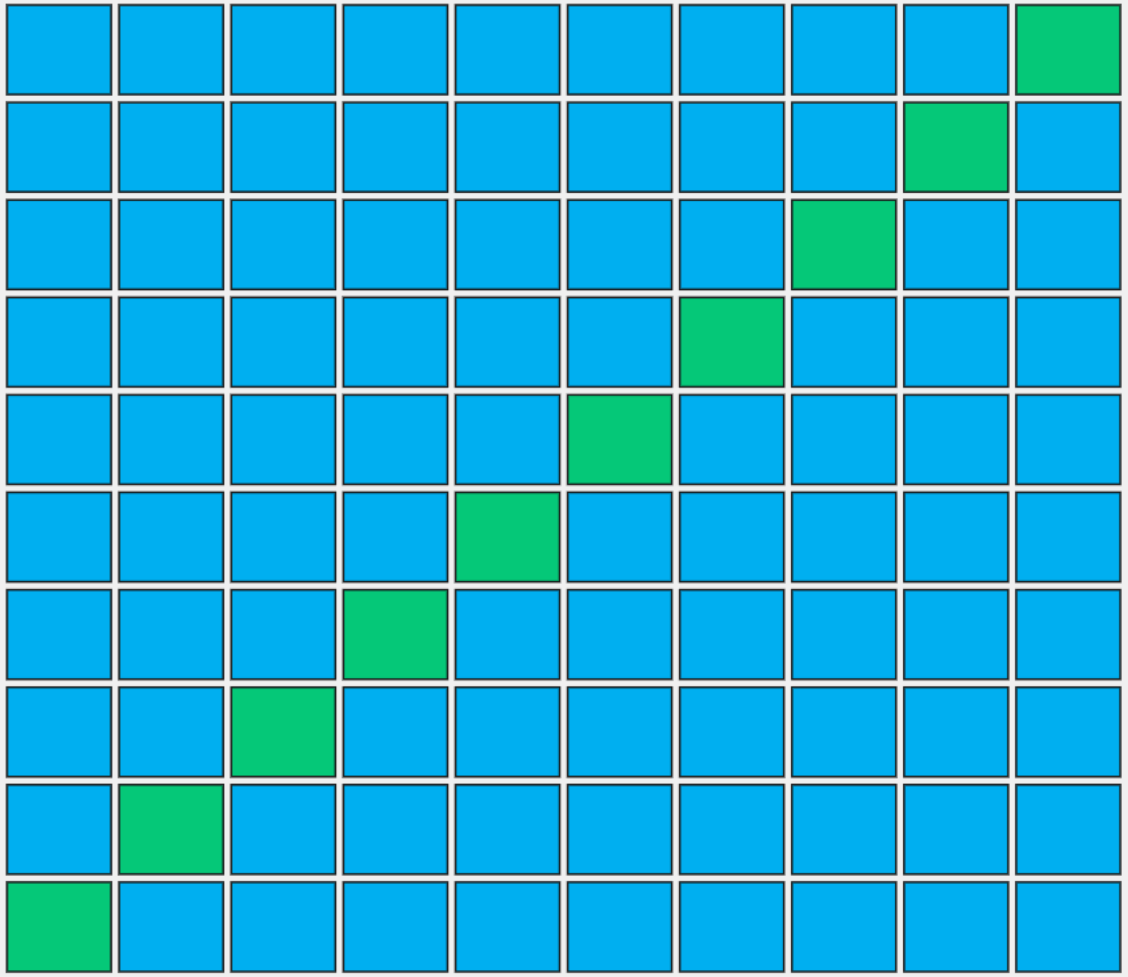
K-fold cross-validation



Test data

The overall process looks something like this. The average of the held-out folds is then taken as the estimate of how well this hyper-parameter setting performs

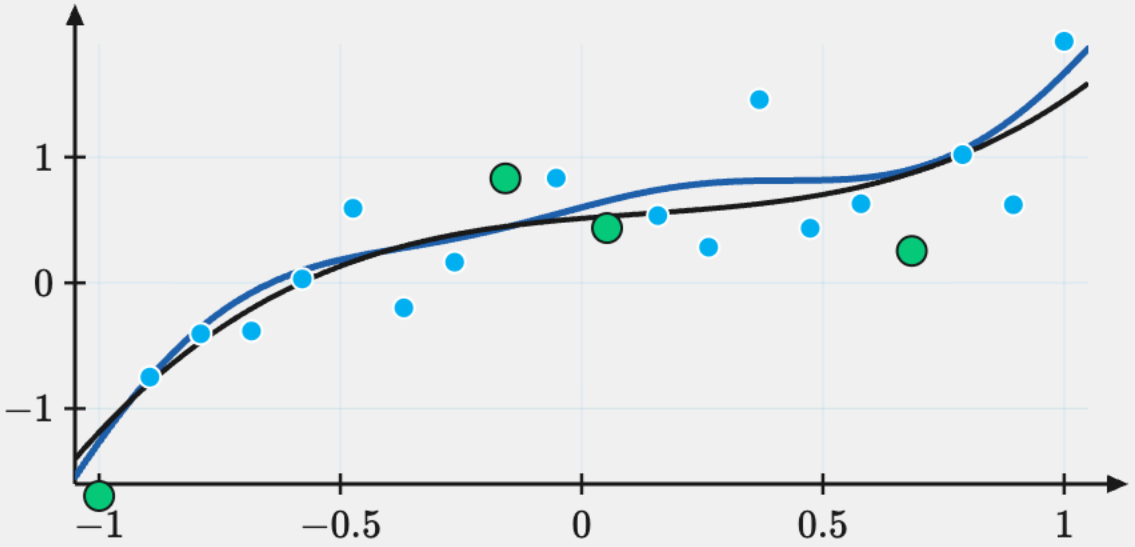
K-fold cross-validation



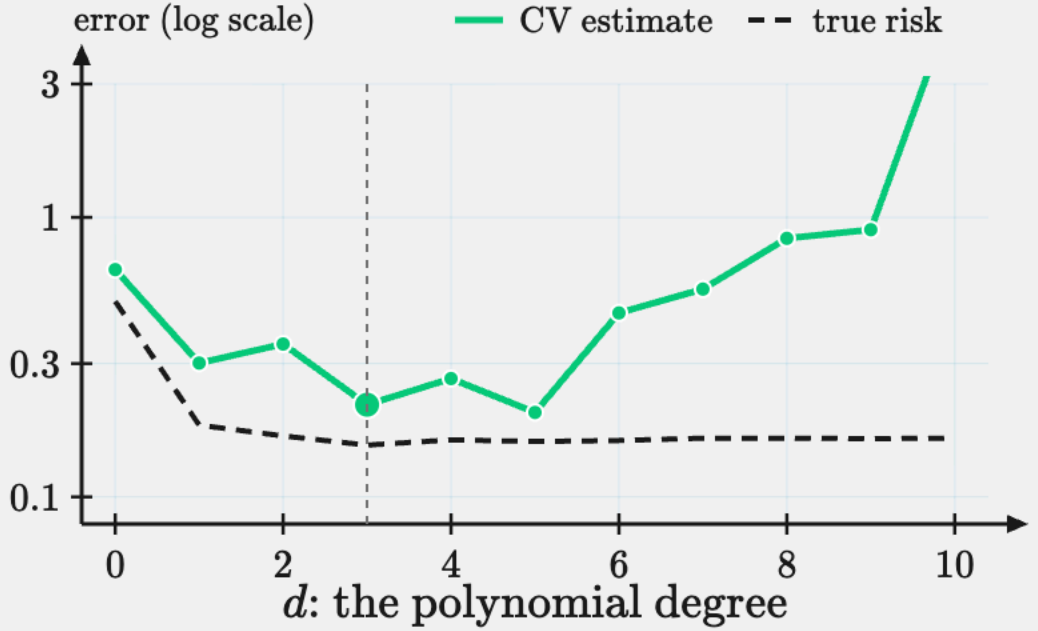
Test data

The most extreme case of k-fold cross-validation is take the number of folds to be equal to the number of data-points which gives us leave-one-out cross validation

K-fold cross-validation, live



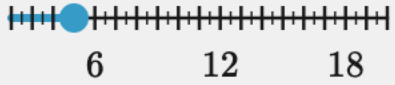




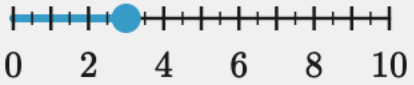
5 folds of 4 points

CV picks $d = 5$; true risk at $d = 3$: 0.153

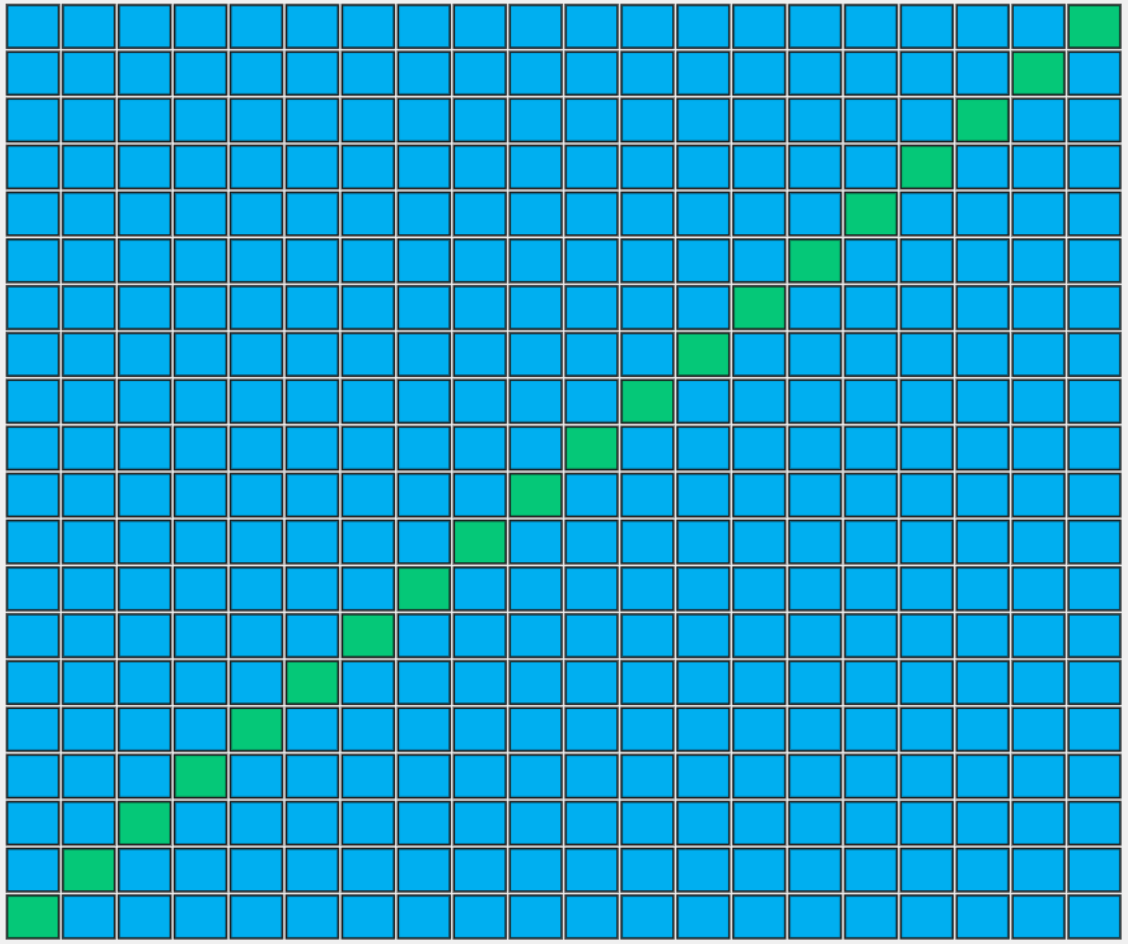
$K = 5$



$d = 3$



K-fold cross-validation



Test data

But does leave-one-out give us a risk estimate that is accurate in the same way that our test-data estimate is?

To think about this, we need to turn to statistics.

Statistics

Definition 7.4.1 (Statistic). A statistic $\mathbf{s}$ is a random variable that is a function of some data $\mathcal{D}$, $\mathbf{s} = g(\mathcal{D})$, where the data $\mathcal{D}$ is a collection of random variables.

Generally, an ‘estimator’ is a statistic that attempts to quantify a particular parameter or property of a distribution

Statistics and estimators

Definition 7.4.1 (Statistic). A statistic $\mathbf{s}$ is a random variable that is a function of some data $\mathcal{D}$, $\mathbf{s} = g(\mathcal{D})$, where the data $\mathcal{D}$ is a collection of random variables.

Generally, an 'estimator' is a statistic that attempts to quantify a parameter or property of a distribution. Example:

$$\underbrace{\mu_{\mathbf{r}}}_{\text{the estimand } S} \longrightarrow \underbrace{\hat{r}_k = \frac{1}{k}\sum_{i=1}^{k} r_i}_{\text{the estimator } \hat{S}}$$

Quantifying bias in a statistic

$$\mu_{\mathbf{r}} \longrightarrow \hat{r}_k = \frac{1}{k}\sum_{i=1}^{k} r_i$$

$$\mathrm{Bias}(\hat{S}) = \mathbb{E}_{\hat{S}}(\hat{S}) - S$$

The bias of the estimator is the expected difference between the estimator and the estimand

Quantifying variance in a statistic

$$\mu_{\mathbf{r}} \longrightarrow \hat{r}_k = \frac{1}{k}\sum_{i=1}^{k} r_i$$

$$\mathrm{Var}(\hat{S}) = \mathbb{E}_{\hat{S}}\left[\left(\hat{S} - \mathbb{E}_{\hat{S}}(\hat{S})\right)^2\right]$$

Given that the estimator is a random variable, its variance is defined correspondingly

Aside: linearity of expectation

In our last lecture we began working with expectations and will do much more of this going forward. It is important to state a couple of facts:

$$\mathbb{E}\left[\sum_{i=1}^{k} \mathbf{x}_i\right] = \sum_{i=1}^{k} \mathbb{E}[\mathbf{x}_i]$$

$$\mathbb{E}[a \cdot \mathbf{x}] = a \cdot \mathbb{E}[\mathbf{x}]$$

$$\mathbb{E}[a\,\mathbf{x} + b\,\mathbf{y}] = a\,\mathbb{E}[\mathbf{x}] + b\,\mathbb{E}[\mathbf{y}]$$

for any random variables, independent or not

Decomposing error of an estimator

Taken together, the bias and variance of an estimator can be used to completely decompose the squared error of our estimator:

$$\mathrm{Err}(\hat{S}) = \mathbb{E}_{\hat{S}}\left[(\hat{S} - S)^2\right] = \mathrm{Bias}^2 + \mathrm{Var}$$

Choosing learning algorithms

$$\text{Let } \mu = \mathbb{E}_{\hat{S}}[\hat{S}]$$

$$\mathbb{E}_{\hat{S}}[(\hat{S} - S)^2] = \mathbb{E}_{\hat{S}}\left[\left(\hat{S} - \mu + \mu - S\right)^2\right]$$

We are going to start with a simple trick that is, adding and subtracting off the mean – just a quantity so this is fine!

Choosing learning algorithms

Let $\mu = \mathbb{E}_{\hat{S}}[\hat{S}]$

$$\mathbb{E}_{\hat{S}}[(\hat{S} - S)^2] = \mathbb{E}_{\hat{S}}\left[\left(\hat{S} - \mu + \mu - S\right)^2\right]$$

$$= \mathbb{E}_{\hat{S}}\left[\left((\hat{S} - \mu) + (\mu - S)\right)^2\right]$$

$$= \mathbb{E}_{\hat{S}}\left[(\hat{S} - \mu)^2 + 2(\hat{S} - \mu)(\mu - S) + (\mu - S)^2\right]$$

Here we have arrived at a sum of random variables inside of an expectation, so we can use linearity of expectation!

Choosing learning algorithms

$$= \mathbb{E}_{\hat{S}}\Big[(\hat{S} - \mu)^2 + 2(\hat{S} - \mu)(\mu - S) + (\mu - S)^2\Big]$$

$$= \mathbb{E}_{\hat{S}}[(\hat{S} - \mu)^2] + \mathbb{E}_{\hat{S}}\Big[2(\hat{S} - \mu)(\mu - S)\Big] + \mathbb{E}_{\hat{S}}[(\mu - S)^2]$$

Application of linearity of expectation

Choosing learning algorithms

$$= \mathbb{E}_{\hat{S}}\left[(\hat{S} - \mu)^2 + 2(\hat{S} - \mu)(\mu - S) + (\mu - S)^2\right]$$

$$= \mathbb{E}_{\hat{S}}[(\hat{S} - \mu)^2] + \mathbb{E}_{\hat{S}}\left[2(\hat{S} - \mu)(\mu - S)\right] + \mathbb{E}_{\hat{S}}[(\mu - S)^2]$$

Application of linearity of expectation + elimination based on linearity of expectation (again):

$\mathbb{E}_{\hat{S}}[2(\hat{S} - \mu)(\mu - S)] = 2(\mu - S)(\mathbb{E}_{\hat{S}}[\hat{S}] - \mu) = 0$, as $\mu - S$ is a number and $\mathbb{E}_{\hat{S}}[\hat{S}] = \mu$

Choosing learning algorithms

$$= \mathbb{E}_{\hat{S}}[(\hat{S} - \mu)^2] + \mathbb{E}_{\hat{S}}\left[2(\hat{S} - \mu)(\mu - S)\right] + \mathbb{E}_{\hat{S}}[(\mu - S)^2]$$

Nothing in this expectation depends on the randomness, so we can remove the expectation.

Choosing learning algorithms

$$\mathbb{E}_{\hat{S}}[(\hat{S} - S)^2] = \mathbb{E}_{\hat{S}}[(\hat{S} - \mu)^2] + (\mu - S)^2$$

$$\mathrm{Var}(\hat{S}) = \mathbb{E}_{\hat{S}}\left[\left(\hat{S} - \mathbb{E}_{\hat{S}}(\hat{S})\right)^2\right] \qquad \mathrm{Bias}(\hat{S}) = \mathbb{E}_{\hat{S}}(\hat{S}) - S$$

After elimination we have arrived at what we wanted!

Choosing learning algorithms

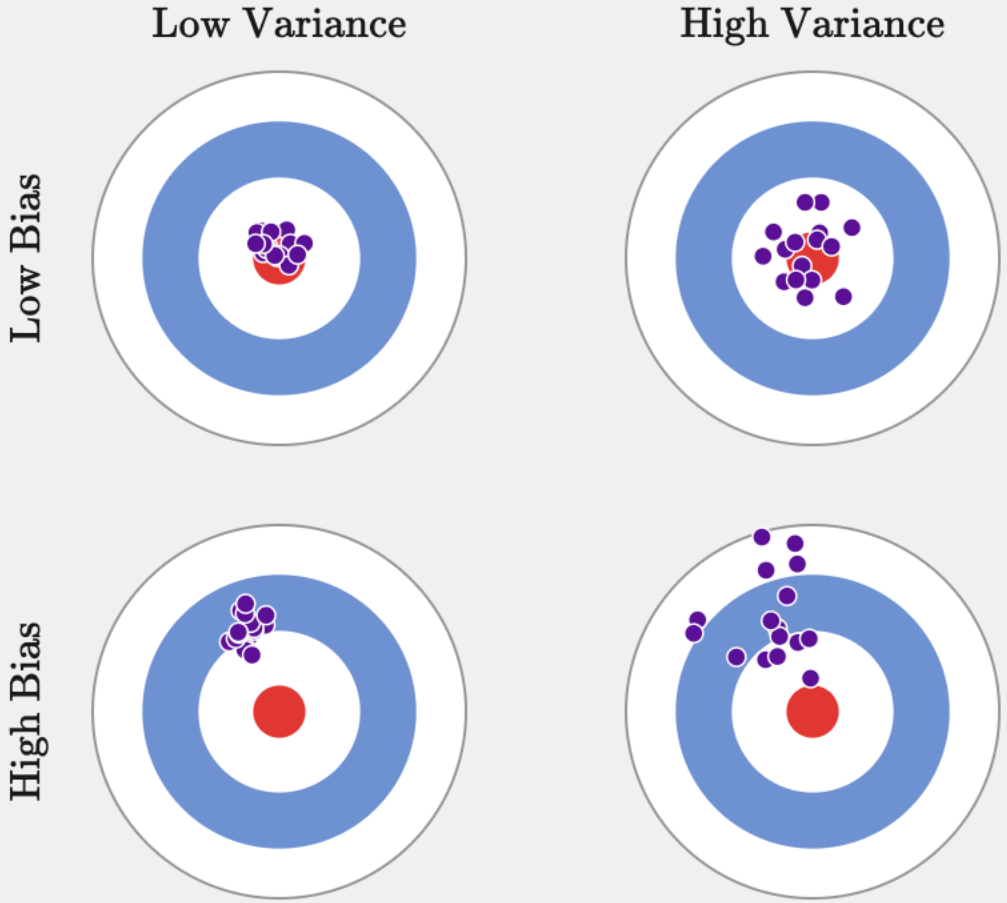
$$\mathbb{E}_{\hat{S}}[(\hat{S} - S)^2] = \underbrace{\mathbb{E}_{\hat{S}}[(\hat{S} - \mu)^2]}_{\text{Var}(\hat{S})} + \underbrace{(\mu - S)^2}_{\text{Bias}(\hat{S})^2}$$

$$\text{Var}(\hat{S}) = \mathbb{E}_{\hat{S}}\left[\left(\hat{S} - \mathbb{E}_{\hat{S}}(\hat{S})\right)^2\right]$$

$$\text{Bias}(\hat{S}) = \mathbb{E}_{\hat{S}}(\hat{S}) - S$$

After elimination we have arrived at what we wanted!

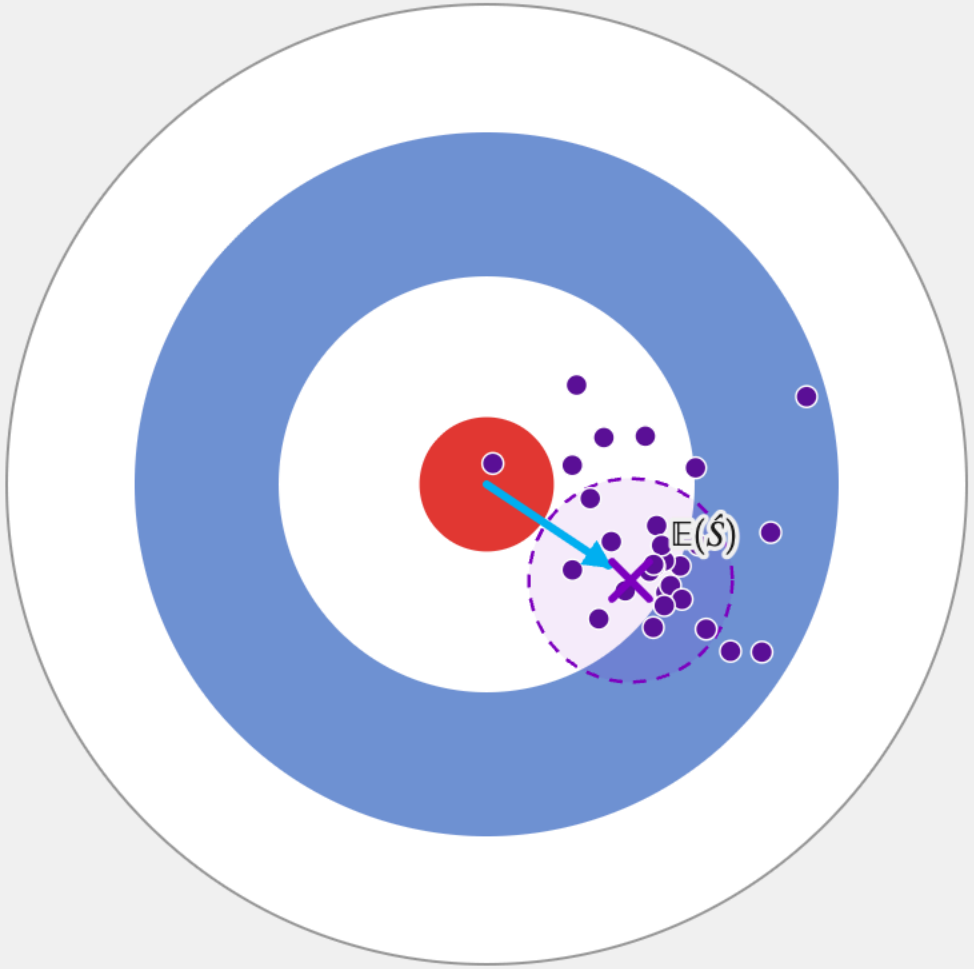
Visualizing bias and variance



$$\mathrm{Var}(\hat{S}) = \mathbb{E}_{\hat{S}}\left[\left(\hat{S} - \mathbb{E}_{\hat{S}}(\hat{S})\right)^2\right]$$

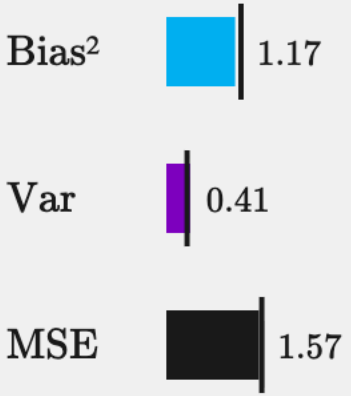
$$\mathrm{Bias}(\hat{S}) = \mathbb{E}_{\hat{S}}(\hat{S}) - S$$

Bias and variance, live



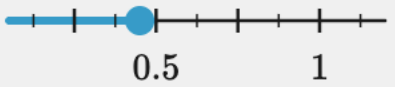
centre: the estimand S

Mean squared error $=$ Bias^2 $+$ Var

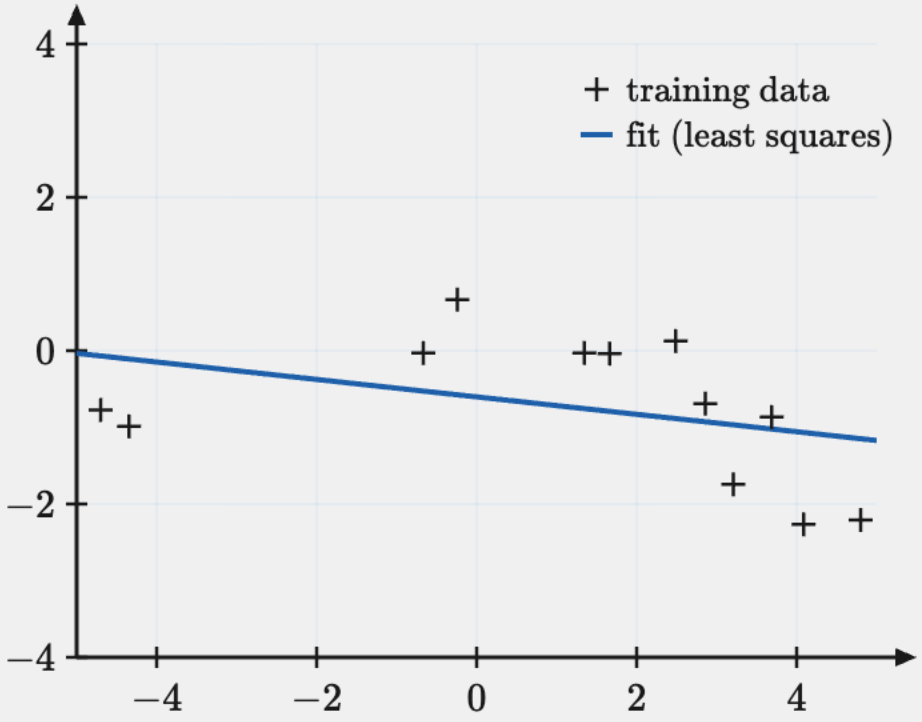


30 estimates: their averages are the black ticks

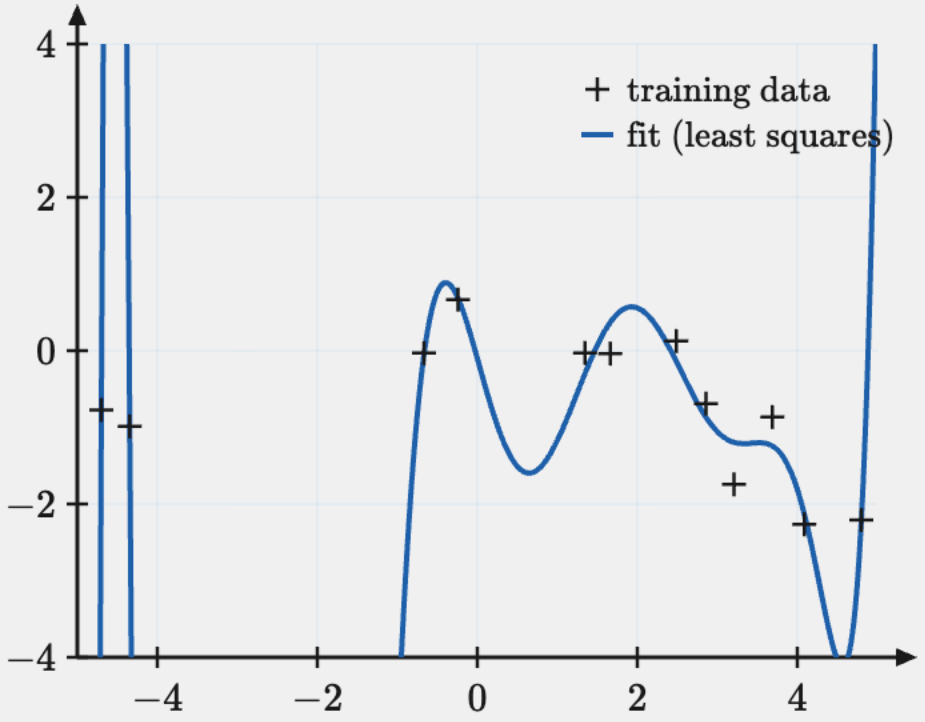
$\sigma = 0.45$



Using this during learning?



$$\phi(x) = [1 \;\; x]^\top$$



$$\phi(x) = [1 \;\; x \;\; x^2 \;\; \cdots \;\; x^k]^\top$$

Bias-variance of predictions

$$\mathbb{E}_{\mathcal{D}}\left[\left(f_{\mathcal{D}}^{\theta}(x) - y\right)^2\right]$$

Let us now think of our machine learning model's prediction at a single point ***x*** to be an estimator of the ground truth value ***y***

NB: *x* and *y* above are representative of a single prediction and are NOT random, but the dataset is!

Bias-variance of predictions

$$\mathbb{E}_{\mathcal{D}}\left[\left(f_{\mathcal{D}}^{\theta}(x) - y\right)^2\right]$$

$$\text{Let } \bar{f}(x) := \mathbb{E}_{\mathcal{D}}\left[f_{\mathcal{D}}^{\theta}(x)\right]$$

As before, we add and subtract off this mean to get:

$$\mathbb{E}_{\mathcal{D}}\left[\left(f_{\mathcal{D}}^{\theta}(x) - y\right)^2\right] = \mathbb{E}_{\mathcal{D}}\left[\left(f_{\mathcal{D}}^{\theta}(x) - \bar{f}(x) + \bar{f}(x) - y\right)^2\right]$$

Bias-variance of predictions

$$\mathbb{E}_{\mathcal{D}}\left[\left(f_{\mathcal{D}}^{\theta}(x) - y\right)^2\right]$$

$$\text{Let } \bar{f}(x) := \mathbb{E}_{\mathcal{D}}\left[f_{\mathcal{D}}^{\theta}(x)\right]$$

And then we get the following decomposition (as expected):

$$= \mathbb{E}_{\mathcal{D}}\left[\left(f_{\mathcal{D}}^{\theta}(x) - \bar{f}(x)\right)^2\right] + \left(\bar{f}(x) - y\right)^2$$

expanding the square: the cross term vanishes, as $\mathbb{E}_{\mathcal{D}}[f_{\mathcal{D}}(x) - \bar{f}(x)] = 0$

Choosing learning algorithms

$$\mathbb{E}_{\mathcal{D}}\left[\left(f_{\mathcal{D}}^{\theta}(x) - y\right)^2\right]$$

$$\text{Let } \bar{f}(x) := \mathbb{E}_{\mathcal{D}}\left[f_{\mathcal{D}}^{\theta}(x)\right]$$

We *could* actually follow a similar break-down pattern of expanding the square on our bias term:

$$= \mathbb{E}_{\mathcal{D}}\left[\left(f_{\mathcal{D}}^{\theta}(x) - \bar{f}(x)\right)^2\right] + \underbrace{\left(\bar{f}(x) - y\right)^2}$$

Choosing learning algorithms

$$= \mathbb{E}_{\mathcal{D}}\left[(f_{\mathcal{D}}^{\theta}(x) - \bar{f}(x))^2\right] + \underbrace{(\bar{f}(x) - y)^2}$$

But neither of these are random variables within an expectation!

We have referenced the idea that our label observations are “noisy” and so if we treat our labels ‘*y*’ as having some randomness then we can decompose this term too!

We will start to dive into the implications of this assumption later :)

Choosing learning algorithms

If *y* is a random variable then:

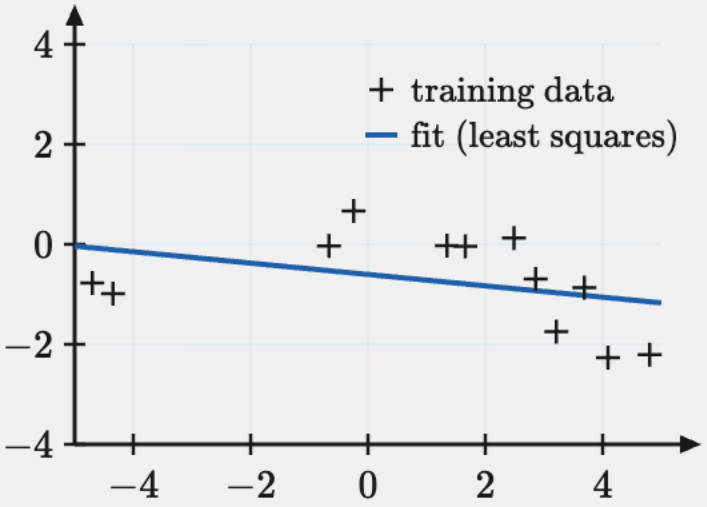
$$\text{Let } \mathbb{E}[y] = \mu_y$$

$$\mathbb{E}_{\mathcal{D}}\left[(f^{\theta}_{\mathcal{D}}(x) - \bar{f}(x))^2\right] + \underbrace{(\bar{f}(x) - y)^2}$$

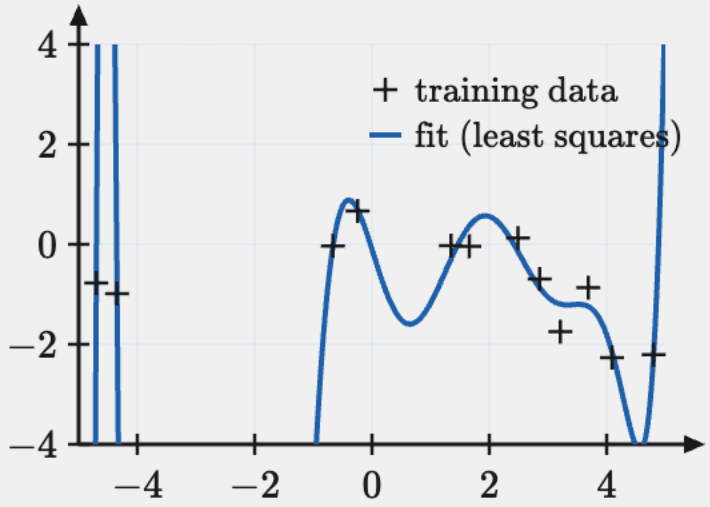
$$= (\bar{f}(x) - \mu_y)^2 + \mathbb{E}\left[(\mu_y - y)^2\right]$$

add and subtract μ_y, expand: the cross term vanishes, as $\mathbb{E}[\mu_y - y] = 0$

Choosing learning algorithms



$$\phi(x) = [1 \;\; x]^\top$$



$$\phi(x) = [1 \;\; x \;\; x^2 \;\; \cdots \;\; x^k]^\top$$

$$\mathbb{E}_{\mathcal{D}}\left[(f^{\theta}_{\mathcal{D}}(x) - y)^2\right] = \underbrace{\mathbb{E}_{\mathcal{D}}\left[(f^{\theta}_{\mathcal{D}}(x) - \bar{f}(x))^2\right]}_{\text{Variance}} + \underbrace{(\bar{f}(x) - \mu_y)^2}_{\text{Squared Bias}} + \underbrace{\mathbb{E}\left[(\mu_y - y)^2\right]}_{\text{Label Noise}}$$

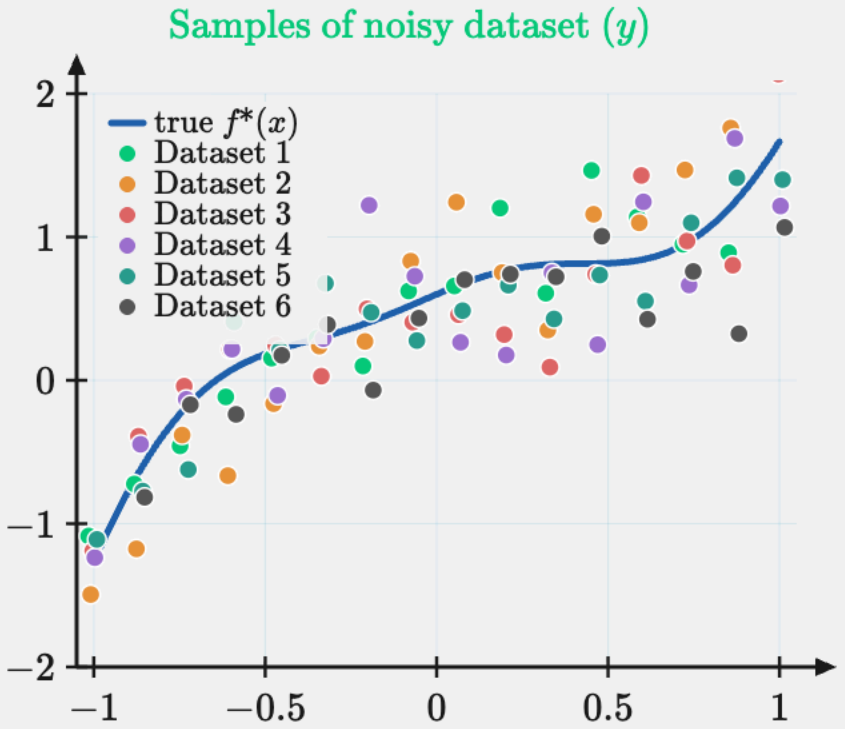
Choosing learning algorithms

$$\mathbb{E}_{\mathcal{D}}\left[(f^{\theta}_{\mathcal{D}}(x) - y)^2\right] = \underbrace{\mathbb{E}_{\mathcal{D}}\left[(f^{\theta}_{\mathcal{D}}(x) - \bar{f}(x))^2\right]}_{\text{Variance}} + \underbrace{(\bar{f}(x) - \mu_y)^2}_{\text{Squared Bias}} + \underbrace{\mathbb{E}\left[(\mu_y - y)^2\right]}_{\text{Label Noise}}$$

This decomposition is very nice because we can begin to think about where our errors come from and then use this insight to understand how we should adjust our models!

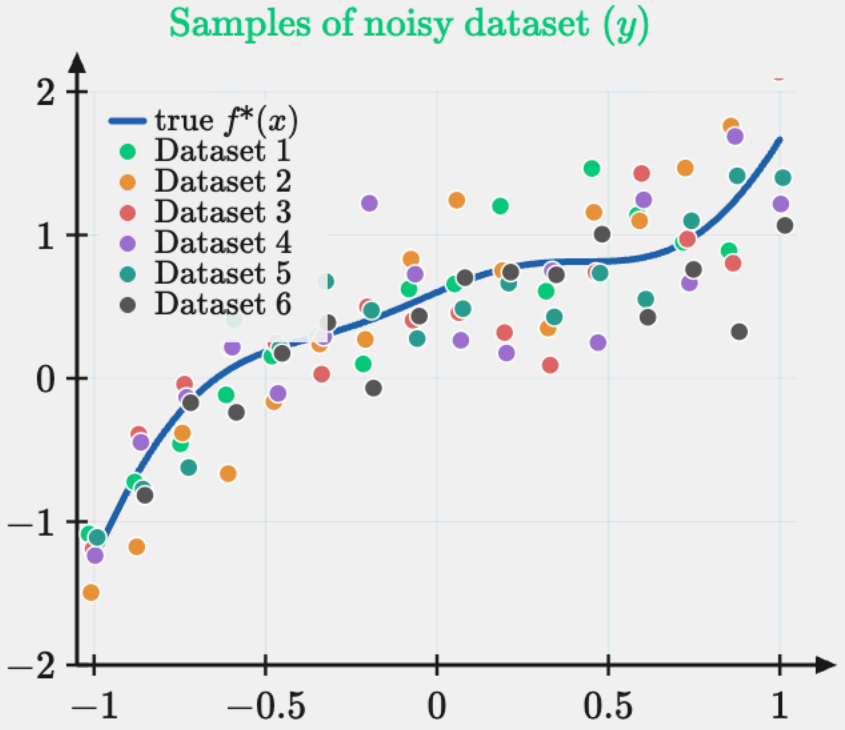
Important: this is for a single prediction!!

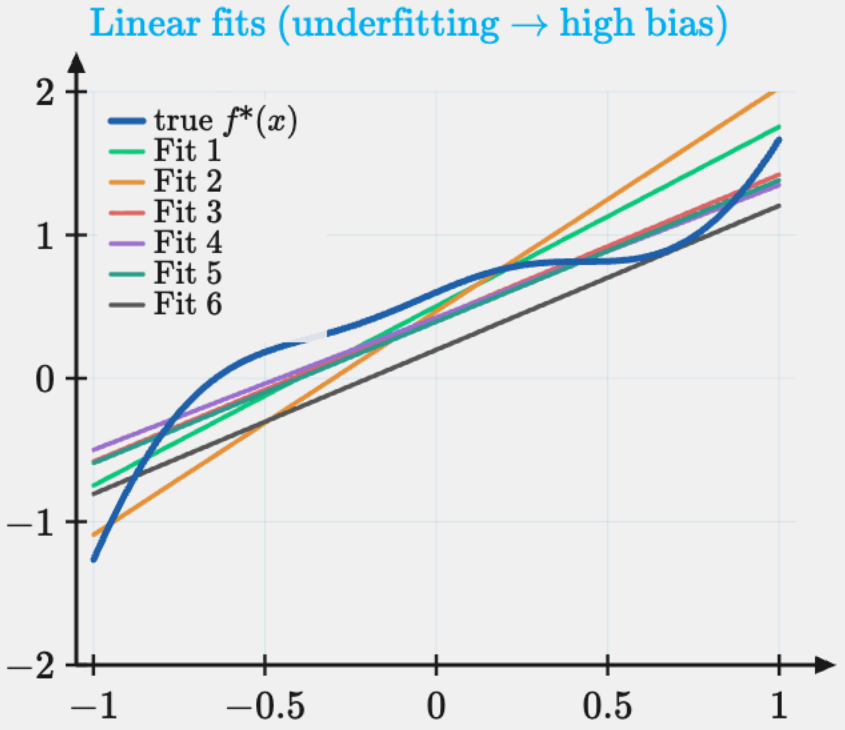
Visualizing bias-variance



$$\mathbb{E}_{\mathcal{D}}\left[(f_{\mathcal{D}}^{\theta}(x) - y)^2\right] = \underbrace{\mathbb{E}_{\mathcal{D}}\left[(f_{\mathcal{D}}^{\theta}(x) - \bar{f}(x))^2\right]}_{\text{Variance}} + \underbrace{(\bar{f}(x) - \mu_y)^2}_{\text{Squared Bias}} + \underbrace{\mathbb{E}\left[(\mu_y - y)^2\right]}_{\text{Label Noise}}$$

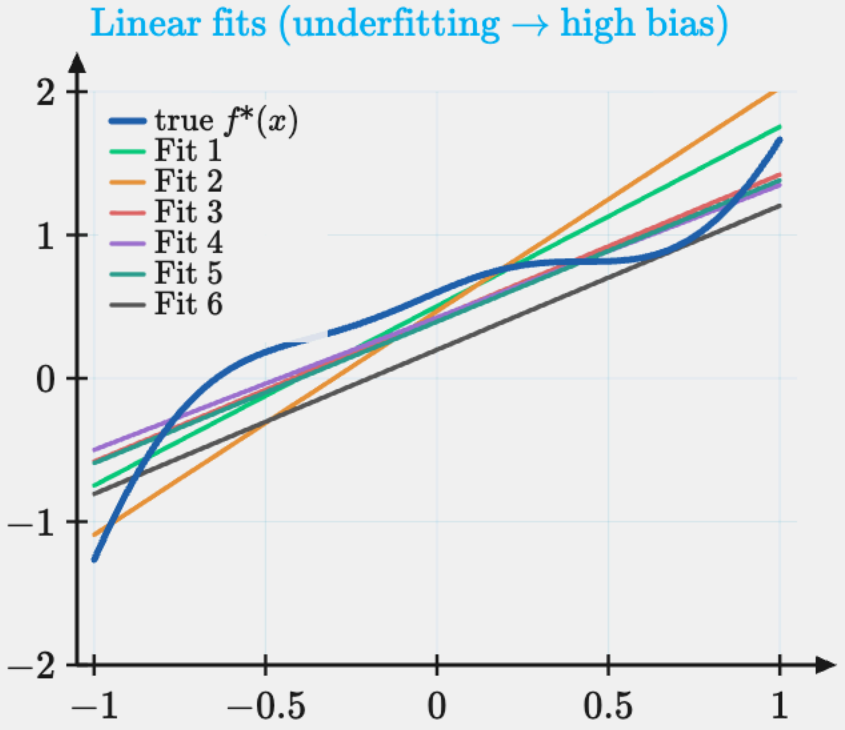
Visualizing bias-variance

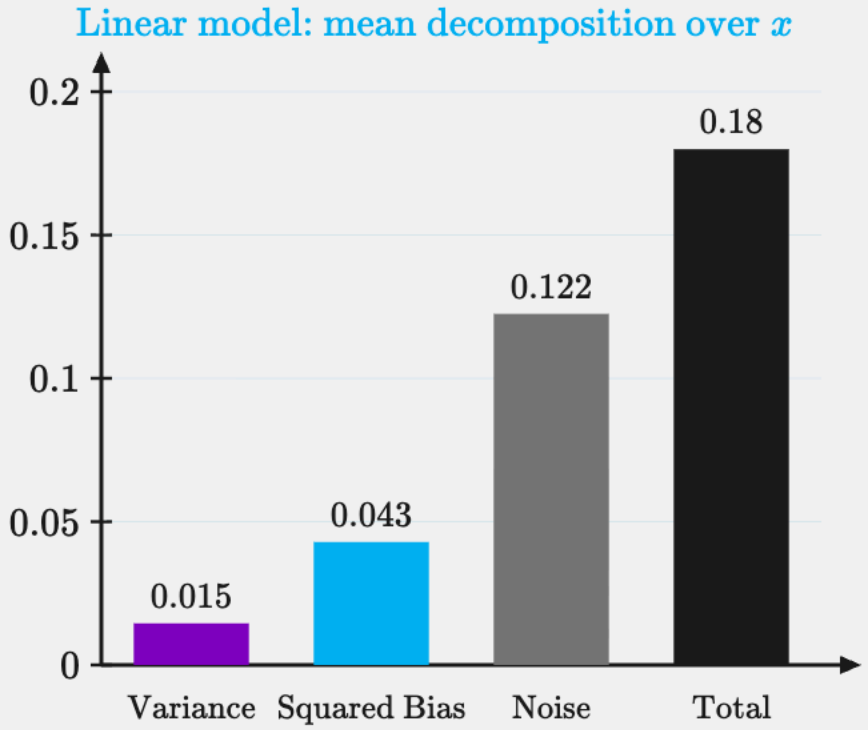




$$\mathbb{E}_{\mathcal{D}}\left[(f_{\mathcal{D}}^{\theta}(x) - y)^2\right] = \underbrace{\mathbb{E}_{\mathcal{D}}\left[(f_{\mathcal{D}}^{\theta}(x) - \bar{f}(x))^2\right]}_{\text{Variance}} + \underbrace{(\bar{f}(x) - \mu_y)^2}_{\text{Squared Bias}} + \underbrace{\mathbb{E}\left[(\mu_y - y)^2\right]}_{\text{Label Noise}}$$

Visualizing bias–variance

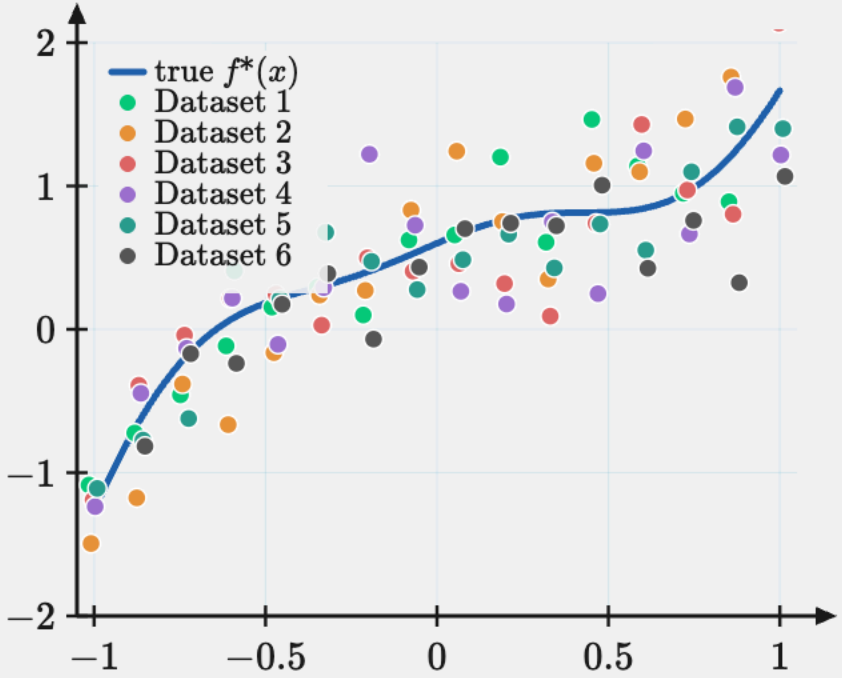




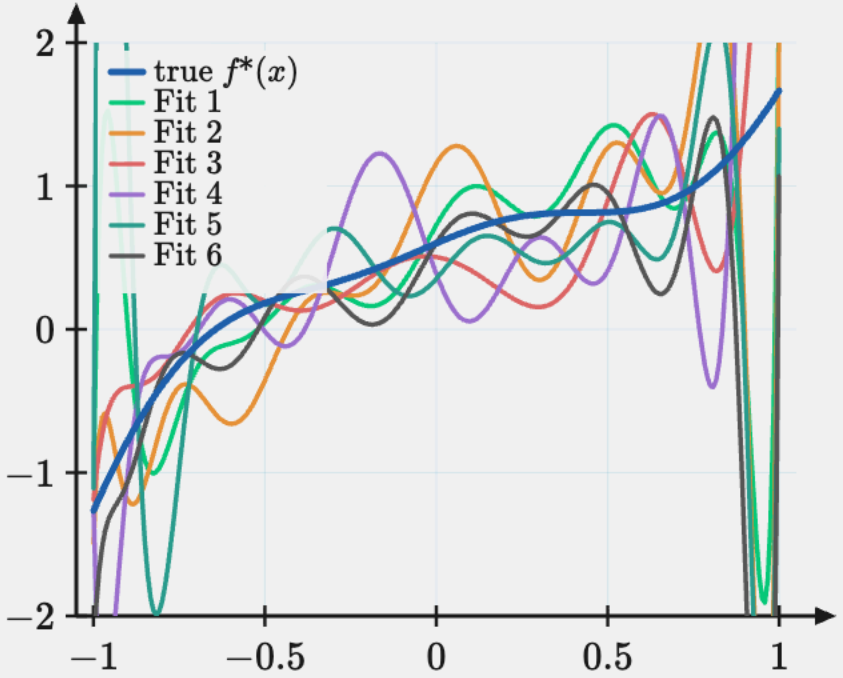
$$\mathbb{E}_{\mathcal{D}}\left[(f_{\mathcal{D}}^{\theta}(x) - y)^2\right] = \underbrace{\mathbb{E}_{\mathcal{D}}\left[(f_{\mathcal{D}}^{\theta}(x) - \bar{f}(x))^2\right]}_{\text{Variance}} + \underbrace{(\bar{f}(x) - \mu_y)^2}_{\text{Squared Bias}} + \underbrace{\mathbb{E}\left[(\mu_y - y)^2\right]}_{\text{Label Noise}}$$

Visualizing bias-variance

Samples of noisy dataset (y)

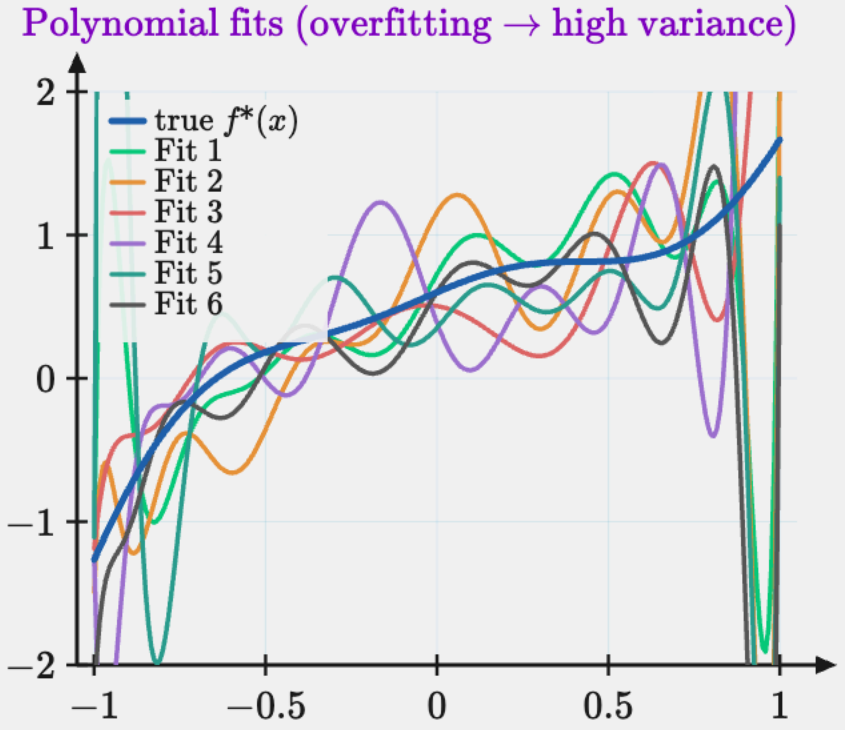


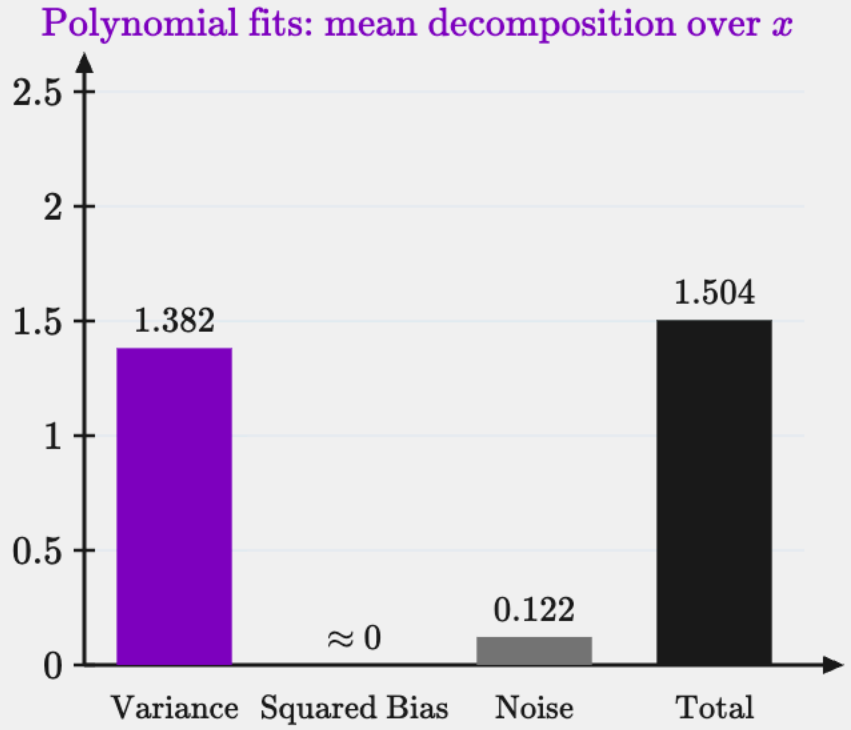
Polynomial fits (overfitting → high variance)



$$\mathbb{E}_{\mathcal{D}}\left[(f_{\mathcal{D}}^{\theta}(x) - y)^2\right] = \underbrace{\mathbb{E}_{\mathcal{D}}\left[(f_{\mathcal{D}}^{\theta}(x) - \bar{f}(x))^2\right]}_{\text{Variance}} + \underbrace{(\bar{f}(x) - \mu_y)^2}_{\text{Squared Bias}} + \underbrace{\mathbb{E}\left[(\mu_y - y)^2\right]}_{\text{Label Noise}}$$

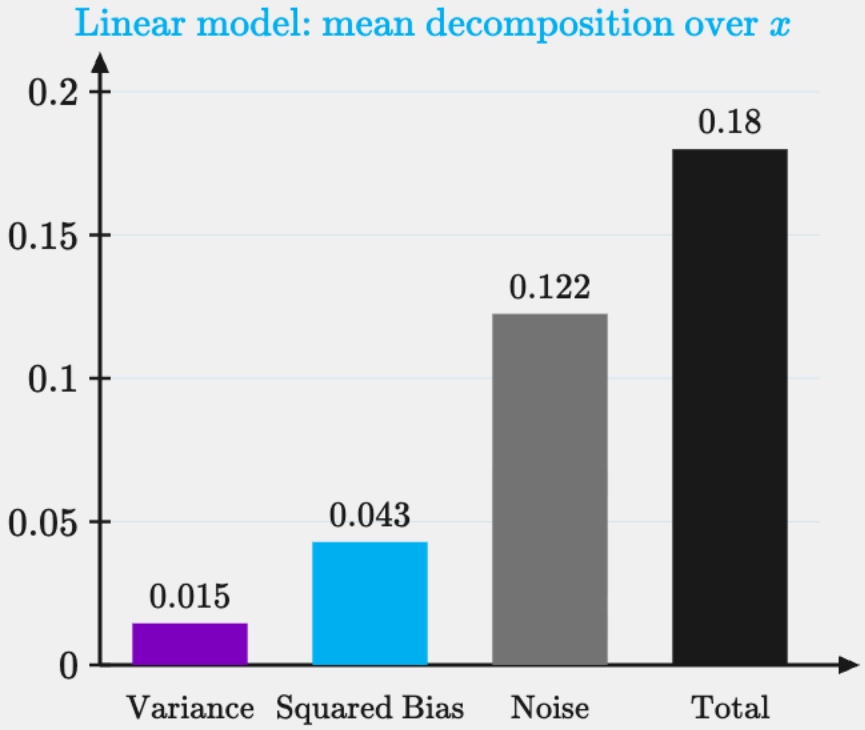
Visualizing bias–variance

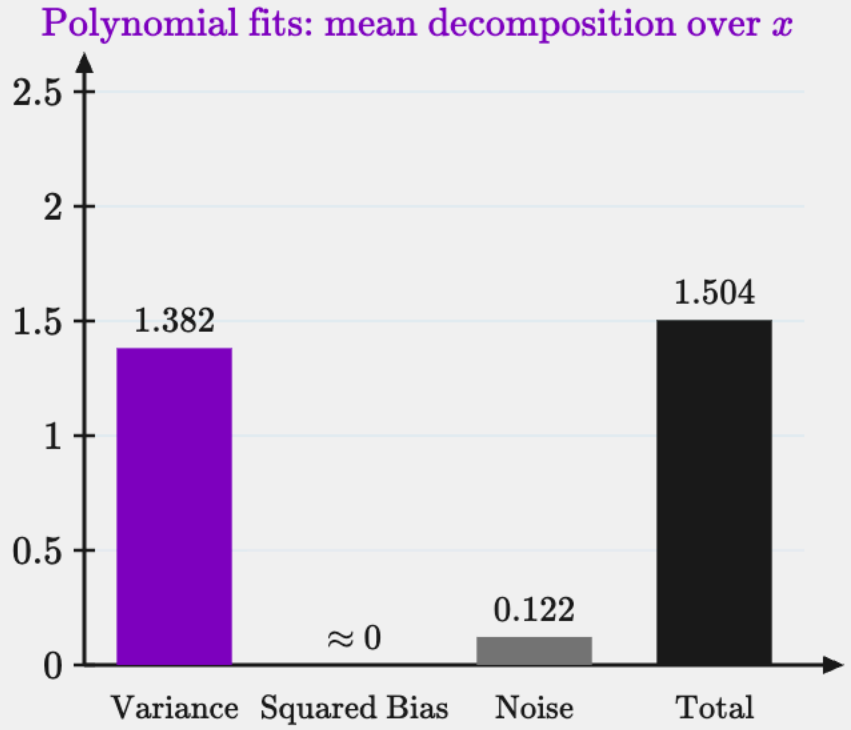




$$\mathbb{E}_{\mathcal{D}}\left[(f_{\mathcal{D}}^{\theta}(x) - y)^2\right] = \underbrace{\mathbb{E}_{\mathcal{D}}\left[(f_{\mathcal{D}}^{\theta}(x) - \bar{f}(x))^2\right]}_{\text{Variance}} + \underbrace{(\bar{f}(x) - \mu_y)^2}_{\text{Squared Bias}} + \underbrace{\mathbb{E}\left[(\mu_y - y)^2\right]}_{\text{Label Noise}}$$

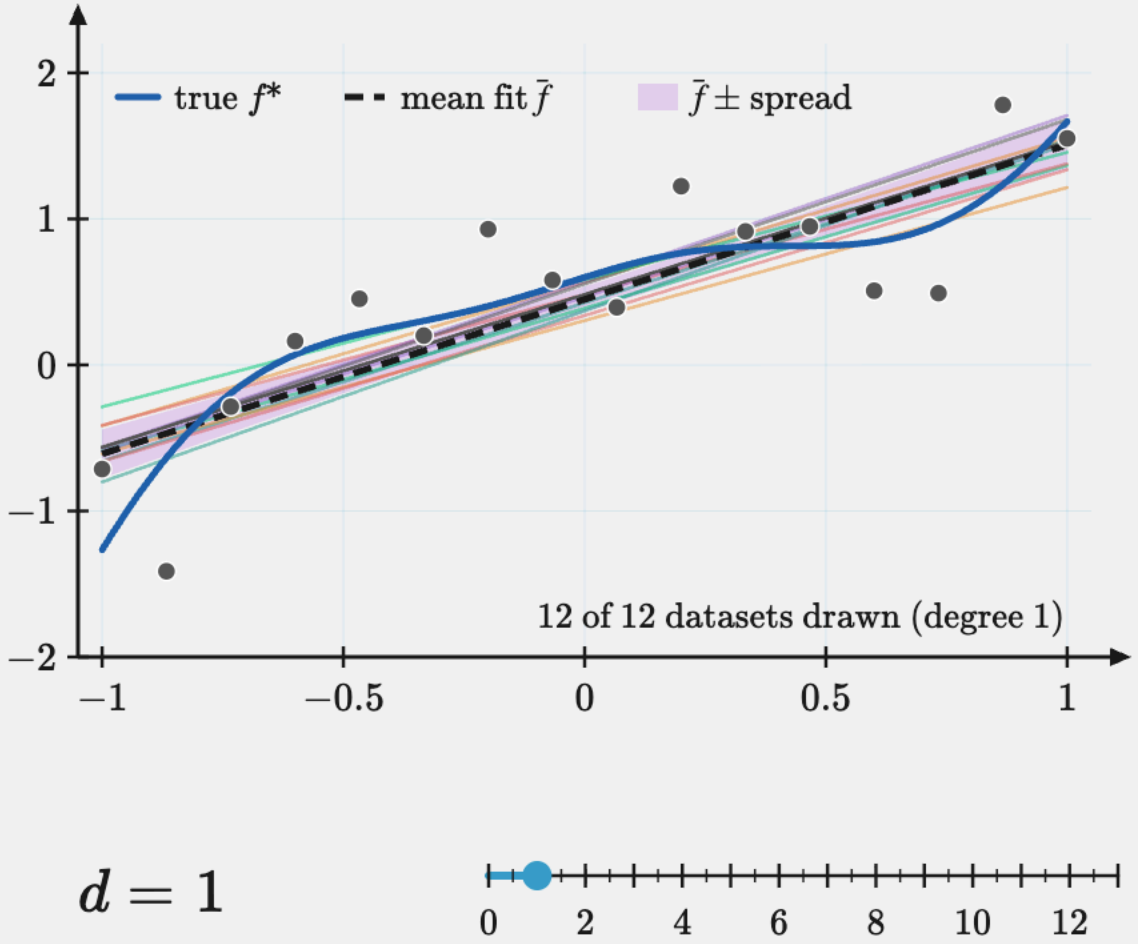
Visualizing bias-variance



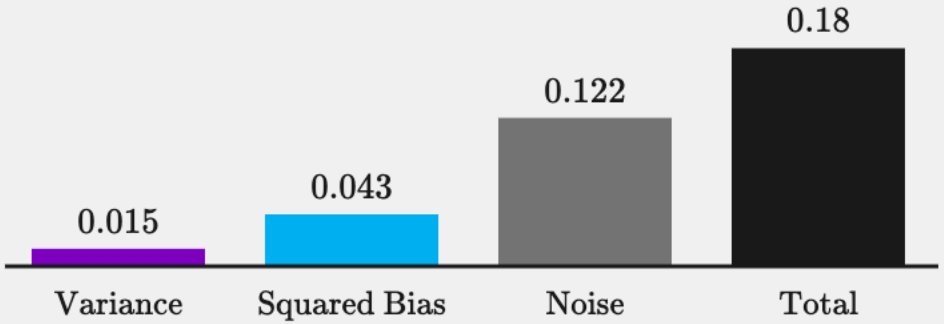


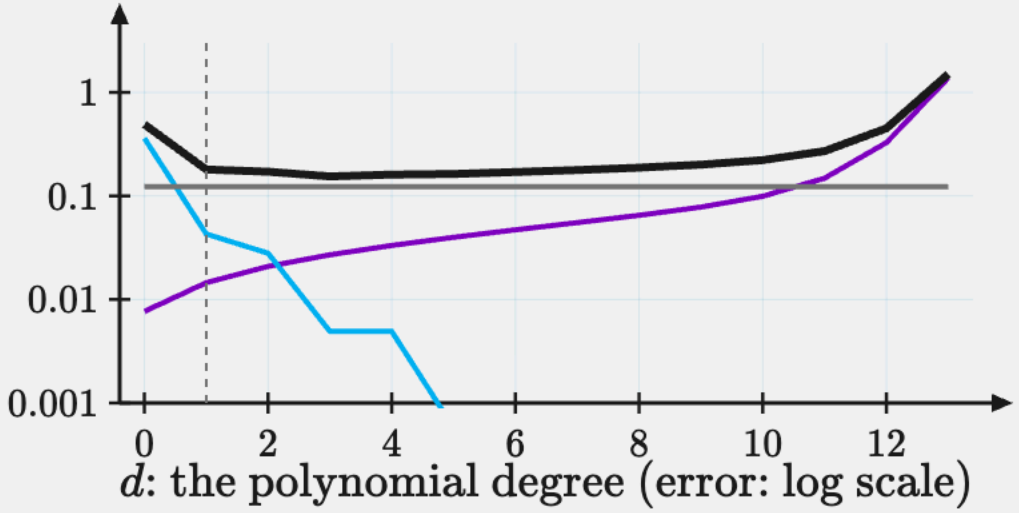
$$\mathbb{E}_{\mathcal{D}}\left[(f_{\mathcal{D}}^{\theta}(x) - y)^2\right] = \underbrace{\mathbb{E}_{\mathcal{D}}\left[(f_{\mathcal{D}}^{\theta}(x) - \bar{f}(x))^2\right]}_{\text{Variance}} + \underbrace{(\bar{f}(x) - \mu_y)^2}_{\text{Squared Bias}} + \underbrace{\mathbb{E}\left[(\mu_y - y)^2\right]}_{\text{Label Noise}}$$

Bias-variance of fits, live



$d = 1$





Bias-Variance in linear models

$$\mathrm{P.Err}(\theta) = \mathbb{E}\left[\|\mathbf{y} - \mathbf{X}\theta\|_2^2\right]$$

$$= \mathbb{E}\left[(\mathbf{X}\theta^\star - \mathbf{X}\theta)^\top (\mathbf{X}\theta^\star - \mathbf{X}\theta)\right]$$

no modelling error: some θ^* generated the data, $\mathbf{y} = \mathbf{X}\theta^*$

Bias-Variance in linear models

$$\mathrm{P.Err}(\theta) = \mathbb{E}\left[\|\mathbf{y} - \mathbf{X}\theta\|_2^2\right]$$

$$= \mathbb{E}\left[(\mathbf{X}\theta^\star - \mathbf{X}\theta)^\top (\mathbf{X}\theta^\star - \mathbf{X}\theta)\right]$$

$$= \mathbb{E}\left[(\mathbf{X}\,\mathrm{Err}(\theta))^\top (\mathbf{X}\,\mathrm{Err}(\theta))\right]$$

writing $\mathrm{Err}(\theta) = \theta^* - \theta$

Bias-Variance in linear models

$$\mathrm{P.Err}(\theta) = \mathbb{E}\left[\|\mathbf{y} - \mathbf{X}\theta\|_2^2\right]$$

$$= \mathbb{E}\left[(\mathbf{X}\theta^\star - \mathbf{X}\theta)^\top (\mathbf{X}\theta^\star - \mathbf{X}\theta)\right]$$

$$= \mathbb{E}\left[(\mathbf{X}\,\mathrm{Err}(\theta))^\top (\mathbf{X}\,\mathrm{Err}(\theta))\right]$$

If the expected error of an estimator is 0, $\mathbb{E}[\mathrm{Err}(\theta)] = 0$, then we say it is unbiased!

Bias-Variance in linear models

We can show that the OLS is unbiased *if* the ground truth can be described as a line (theta star):

$$\theta^{\mathrm{OLS}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$$

$$= (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top (\mathbf{X}\theta^\star)$$

$$\mathbb{E}\left[(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top (\mathbf{X}\theta^\star)\right] = \theta^\star$$

$(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{X} = \mathbf{I}$: the inverse cancels the term multiplying the parameter

Bias-Variance in linear models

We can show that the Ridge is biased:

$$\theta^{\text{Ridge}} = (\lambda \mathbf{I} + \mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$$

$$= (\lambda \mathbf{I} + \mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{X} \theta^\star$$

$$\mathbb{E}\left[(\lambda \mathbf{I} + \mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top (\mathbf{X}\theta^\star)\right] = \mathbb{E}\left[(\lambda \mathbf{I} + \mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{X}\right] \theta^\star$$

The algebra here is straight-forward but non-trivial so I will reference you to the notes to follow along with it!

Bias-Variance in linear models

We can show that the Ridge is biased:

$$\mathbb{E}_{\mathcal{D}}[\theta^{\text{Ridge}}] - \theta^\star = \mathbb{E}\left[(\lambda\mathbf{I} + \mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{X}\right]\theta^\star - \theta^\star$$

$$= -\lambda\,\mathbb{E}\left[(\lambda\mathbf{I} + \mathbf{X}^\top\mathbf{X})^{-1}\right]\theta^\star$$

as $(\lambda\mathbf{I} + \mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{X} = \mathbf{I} - \lambda(\lambda\mathbf{I} + \mathbf{X}^\top\mathbf{X})^{-1}$

Recover the OLS for $\lambda = 0$, though

Exam question presentation